

Plateaus and Jumps: A Dual-Stream Attention Agent on Held-Frame Change Detection

How a parallel salience / top-down recurrent vision transformer solves cued, value-weighted change detection when each frame is held for five steps

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Abstract

We analyse a parallel dual-stream recurrent vision transformer trained by reinforcement to perform cued, value-weighted change detection under a held-frame protocol: each of seven logical frames is presented for five identical physical steps, so the agent observes a thirty-five step video in which the orientation change appears at physical step twenty-five. The architecture carries two recurrent memories read by two parallel attention streams — a salience stream whose queries are the percept and whose keys/values are the first memory, and a top-down stream keyed on the second memory — whose outputs are concatenated and read by convolutional actor and critic heads. Holding each frame turns the agent’s internal dynamics into a sequence of plateaus and jumps, which we use to read out its computation frame by frame. The trained agent is a conservative, well-calibrated detector: it withholds almost perfectly on no-change trials and detects supra-threshold changes graded by magnitude, with a psychometric threshold near a quarter of the maximum orientation change. Its behaviour is value-directed — it abandons the lowest-value cue colour — but, as in the un-repeated task, it shows no spatial cueing benefit. Faithful attention extraction, linear decoding, and causal bias injection together show a clean division of labour: the salience stream holds a grounded, cue-shaped read of the scene that prevents collapse, while the top-down stream is a change-triggered gate that, when biased onto the changed location, is the sole strong causal lever on the decision. The critic carries a near-binary, value-graded code whose uncertainty falls with the strength of the evidence.

1. Introduction

A central question for models of visual attention is how an agent that is rewarded only sparsely — for a single correct report at the end of a trial — comes to allocate internal attention in space and time, and how that allocation relates to the decision it ultimately makes. We study this in a recurrent vision transformer trained end-to-end by reinforcement on a Posner-style cued change-detection task, in a held-frame variant designed to expose the agent’s temporal dynamics.

In the standard task each video frame is a fresh draw of oriented Gabor patches plus pixel noise, so the agent must integrate evidence across frames that are individually noisy. Here we instead hold each logical frame for five physical steps: the pixels (including the noise) are frozen within a logical frame, so five consecutive observations are identical, and a new noise draw occurs only when the logical frame advances. The agent therefore sees a thirty-five step episode built from seven logical frames, with the cue present from the start and the orientation change appearing at logical frame five — physical step twenty-five. This manipulation does two things. First, it lengthens the agent’s look at each image, which empirically lets this architecture learn a task it otherwise struggles with. Second, and more useful for analysis, it turns the recurrent state’s trajectory into a staircase: within a held frame the input is constant, so the state relaxes onto a plateau; when the frame advances — and especially when the change arrives — the state jumps. Reading the agent on this staircase lets us separate what is driven by fresh sensory evidence from what is intrinsic recurrent dynamics, and aligns naturally with a view of the latent as a point that drifts on plateaus and jumps between basins.

We ask five questions, matching the canonical deep-dive battery for this architecture. (i) What is the agent’s behavioural signature — its psychometric and chronometric functions, its sensitivity to the cue’s stated reliability, and its dependence on cue value? (ii) How is attention allocated across the two streams over the held-frame timeline? (iii) What do the two recurrent memories encode, and when? (iv) Which stream causally carries the decision, and which carries value? (v) What does the distributional critic represent? We answer each with faithful, intervention-ready measurements rather than proxies.

2. Methods

2.1 The held-frame change-detection environment

Each trial presents four oriented Gabor patches, one per quadrant (S1 top-left through S4 bottom-right), on a noisy background, with a coloured spatial cue ring whose radius encodes a stated reliability (proportion $\in \{0.25, 0.5, 0.75, 1.0\}$) and whose colour (red, green, blue) sets the trial’s reward value (5, 3, 1). On a change trial ($p = 0.5$) one patch rotates by $\Delta\theta$ degrees, drawn during training from $\text{Unif}(-64^\circ, 64^\circ)$; on a no-change trial nothing rotates. The agent emits wait or press at each physical step. Pressing on or after the change frame of a change trial yields the cue-colour reward (a hit); pressing before the change, or on a no-change trial, ends the trial with zero reward (a false alarm); waiting out a no-change trial to the end also yields the cue-colour reward (a correct rejection). Overall accuracy is therefore the average of the hit rate on change trials and the correct-rejection rate on no-change trials.

The held-frame protocol holds each of seven logical frames for five physical steps with frozen pixels, giving a thirty-five step episode with the change at physical step twenty-five. All analyses below are aligned to this physical timeline; reaction time is measured in physical steps after the change onset, capturing within-frame deliberation.

2.2 Architecture

The agent is conv-free. Each frame is cut into a 10×10 grid of $5 \times 5 \times 3$ patches, each linearly embedded to a $d = 128$ token. A parallel dual-stream encoder carries two per-token recurrent memories H_1, H_2 and runs two attention blocks each step:

- Saliency (bottom-up): $Z_{\text{sal}} = X + \text{attn}(Q=X, K=V=H_1) + \text{FFN}$, residual on the percept X ;
- Top-down (gating): $Z_{\text{td}} = H_2 + \text{attn}(Q=X, K=V=H_2) + \text{FFN}$, residual on the second memory H_2 .

The memories are then updated by per-token LSTM cells, $H_1 \leftarrow \text{LSTM}_1(X)$ and $H_2 \leftarrow \text{LSTM}_2(Z_{\text{sal}})$. The two stream outputs are concatenated, $[Z_{\text{sal}} \| Z_{\text{td}}]$, and read by 1-D-convolutional actor and critic heads (no CLS token). The critic is distributional (51 quantiles). Training is PAC (MPO + behaviour cloning) with a QR-DQN critic and prioritized episode replay. The model has roughly 1.1M parameters and reaches an overall accuracy near 0.74 on the held-frame task.

2.3 Faithful attention extraction and intervention

All attention analyses use a bias-injectable replica of the encoder step that reproduces the trained model’s actor logits to within 2×10^{-6} , so the maps and interventions are the agent’s own computation, not a re-trained proxy. For each stream we read the pre-softmax attention of the percept queries onto the memory keys, average over heads and queries, and pool the resulting distribution over the four quadrants (it sums to one over the 100 keys). For causal tests we add a scalar bias to the pre-softmax logits of a chosen stream and key region (one quadrant, or all keys uniformly) and sweep it over $[-6, +6]$, measuring the effect on the decision (hit rate) and on value (critic entropy).

2.4 Behavioural, decoding, causal and value analyses

Behaviour uses greedy rollouts under forced trial specifications (fixed cue side, validity, magnitude, colour). Attention, decoding and value use forced-wait rollouts so the full thirty-five-step trajectory is observed. Decoding fits cross-validated linear classifiers to the per-quadrant-pooled memories at each physical step. Reported numbers are over hundreds of trials per condition; the analysis is run serially on CPU.

3. Results

3.1 A conservative, value-directed detector with no spatial cueing benefit

On the natural change/no-change mix the agent reaches an overall accuracy of 0.69, which decomposes into a high correct-rejection rate (0.94), a moderate hit rate (0.46), and a low false-alarm rate (0.06) (Figure 1a). The criterion is conservative: the agent rarely presses on a no-change trial, paying for that caution with missed changes rather than false alarms. Detection is graded by the physical signal — at the fully reliable cue the hit rate climbs from chance at 0° to about 0.60 at the largest orientation change, with a threshold near $16\text{--}28^\circ$, roughly a quarter to a half of the maximum (Figure 1b). Reaction time, measured in physical steps after the change onset, is essentially zero: once the agent detects a change it presses on the first held step of the change frame, so the held protocol buys detection accuracy rather than graded deliberation time.

Two classical attention signatures are then notable by their absence and presence. There is no spatial cueing benefit: the psychometric and chronometric functions for valid and invalid cues are superimposed (at the fully reliable cue, hit rates of 0.60 valid versus 0.62–0.72 invalid across the upper magnitudes — if anything invalid is marginally higher), and hit rate at the largest magnitude is flat across displayed cue reliability (0.60–0.73 from ring 1.0 to 0.25; Figure 1c). In contrast there is a strong value signature (Figure 1d): at a near-threshold magnitude the agent detects high-value changes well (red 0.88, green 0.92) but abandons the lowest-value colour almost entirely (blue 0.00). The agent reads the cue for what the trial is worth, not for where to look.

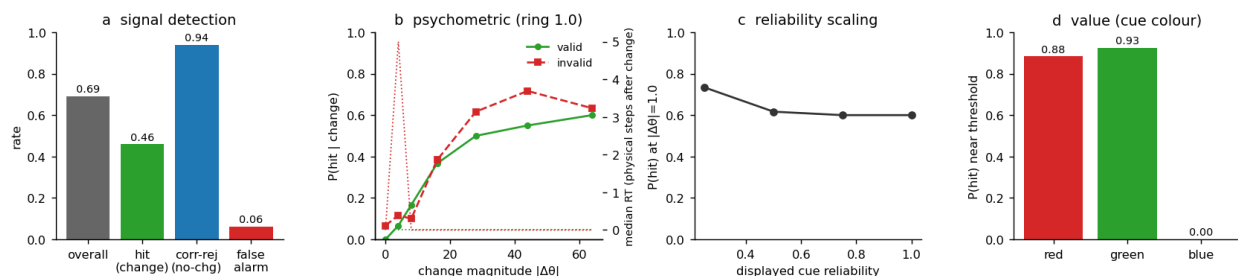


Figure 1: Behaviour of the held-frame agent. (a) Signal-detection decomposition on the natural mix: overall accuracy is the mean of a high correct-rejection rate and a moderate hit rate, with a low false-alarm rate. (b) Psychometric (solid, left axis) and chronometric (dotted, right axis) functions versus orientation-change magnitude for valid and invalid cues — a graded detection function with the valid and invalid curves overlapping. (c) Hit rate at the largest magnitude versus displayed cue reliability — flat. (d) Hit rate near threshold by cue colour — the agent abandons the lowest-value colour.

3.2 Two streams: a grounded salience read and a change-triggered top-down gate

The held-frame timeline makes the two streams’ division of labour visible as plateaus and a jump (Figure 2). On a validly cued, fully reliable trial with the change at S1, the salience stream holds a mildly cue-shaped, roughly stable read of the scene: its attention onto the cued quadrant sits around 0.27 and barely moves through the change (0.27 before, 0.24 after), and across quadrants it is close to uniform with a slight bias toward the cued side (S1 0.27 versus S4 0.20). The top-down stream behaves oppositely: its read of S1 is flat at about 0.31 through the pre-change plateaus and then jumps to 0.44–0.48 within a frame of the change, while its attention to the other three quadrants falls to 0.15–0.20 — a clean change-lock onto the changed location (Figure 2, right; spatial maps in Figure 4).

This lock is universal and cue-independent (Figure 3). Plotting attention onto the changed quadrant across all eight cue conditions, the top-down jump at step twenty-five is present on every condition regardless of validity or displayed reliability, whereas the salience stream’s pre-change read of that quadrant is slightly higher when it is the cued side than when it is not — a weak spatial cue prior carried by the grounded stream, not the gate. The salience stream supplies a stable, scene-grounded read; the top-down stream is a trigger that fires on change.

3.3 What the two memories encode, and when

Linear decoding from the per-quadrant memories shows that the agent’s representation tracks what it uses and discards what it ignores (Figure 5). The cue colour — the trial’s value, which the agent acts on — is decodable at ceiling from the top-down memory throughout the trial (balanced

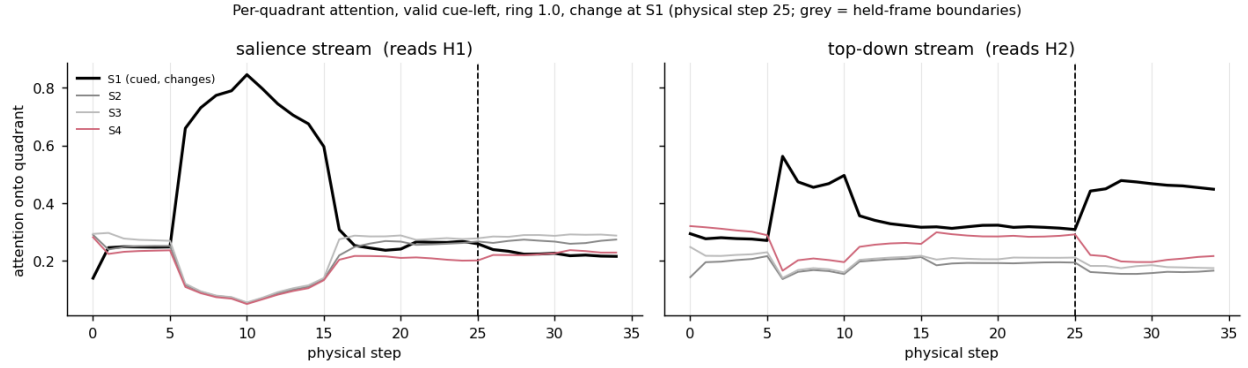


Figure 2: Per-quadrant attention over the held-frame timeline for the canonical valid cue (cue-left, ring 1.0, change at S1 at physical step 25; grey lines mark held-frame boundaries, black dashes mark change onset). The salience stream (left) holds an elevated read of the cued quadrant throughout; the top-down stream (right) is roughly uniform until the change and then snaps onto the changed quadrant.

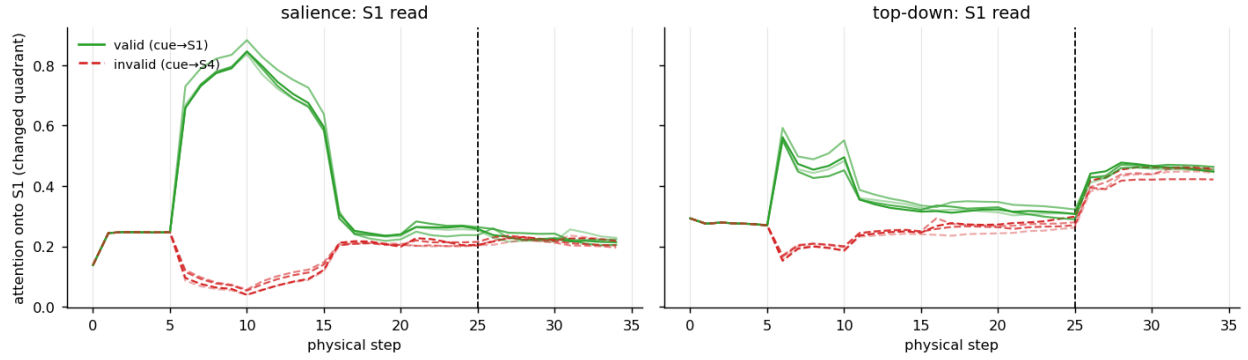


Figure 3: Attention onto the changed quadrant S1 over time for all eight cue conditions (green solid = valid/cue-left, red dashed = invalid/cue-right; opacity = displayed reliability). The top-down change-lock at step 25 is universal; the salience stream's pre-change S1 read is higher on valid than invalid trials.



Figure 4: Spatial attention maps (head- and query-averaged) at five physical timepoints, salience (top) and top-down (bottom); the cyan box marks the cued/changed quadrant S1.

accuracy 1.00 at every step; 0.96 from the salience-fed memory at the end). The cue reliability — which the agent’s behaviour ignores — tells the opposite story: it is perfectly decodable early, while the cue ring is on the screen (1.00 at step ten), but decays to chance (0.25) by the change frame. The agent perceives reliability but does not maintain it.

The changed quadrant is the cleanest temporal signature. With the change location randomised independently of the cue, its decodability sits exactly at chance (0.25–0.26) through every pre-change plateau and then jumps to 0.96 from the top-down memory within a frame of the change (0.71 from the salience-fed memory). The spatial identity of the change is therefore created at the moment of the change and held primarily in the top-down memory — the same memory whose attention locks onto that quadrant in §3.2.

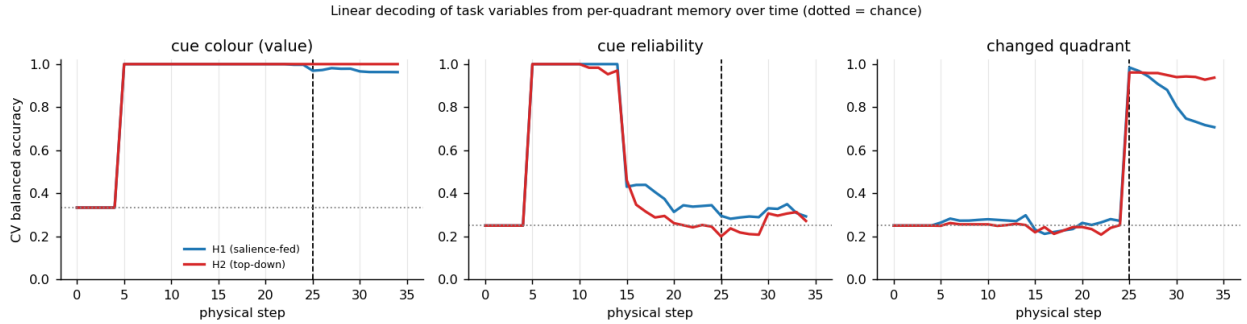


Figure 5: Linear decoding of task variables from the per-quadrant memories over the held-frame timeline (cross-validated balanced accuracy; dotted lines mark chance). Cue colour and reliability are held from cue onset; the changed quadrant comes online only after the change.

3.4 Which stream causally carries the decision

Injecting an attention bias into one stream at a time, and sweeping it over $[-6, +6]$ on near-threshold change-at-S1 trials, separates the two streams by what they control (Figure 6). For the decision (hit rate), the only lever with a non-negative effect is biasing top-down attention onto the changed quadrant: at the strongest positive bias the hit rate rises to about 0.64 from an unbiased 0.54, while uniform gains and both salience levers leave it unchanged or lower it (sweep effects of $+0.05$ for top-down→S1 versus -0.05 , -0.11 , and -0.14 for the others). The decision effect is modest — near the trial-to-trial noise floor — so we read it as directionally consistent with the un-repeated model rather than a large lever.

The dissociation is sharper for value. Biasing the salience stream onto the changed quadrant strongly moves the critic’s outcome uncertainty (entropy change $+0.33$ over the sweep), whereas biasing the top-down stream onto the same quadrant moves the decision but leaves the critic’s uncertainty essentially untouched ($+0.001$). The two streams thus push different variables: top-down attention nudges whether the agent acts, salience attention nudges how certain the value estimate is.

3.5 A calibrated, value-graded critic

Under a forced-wait policy the distributional critic carries a value-graded code (Figure 7). State value before the change is ordered by cue colour in rough proportion to the reward on offer (red 3.8, green 2.2, blue 0.4 for rewards of 5, 3, 1), with no-change trials at an intermediate 2.0 — the value

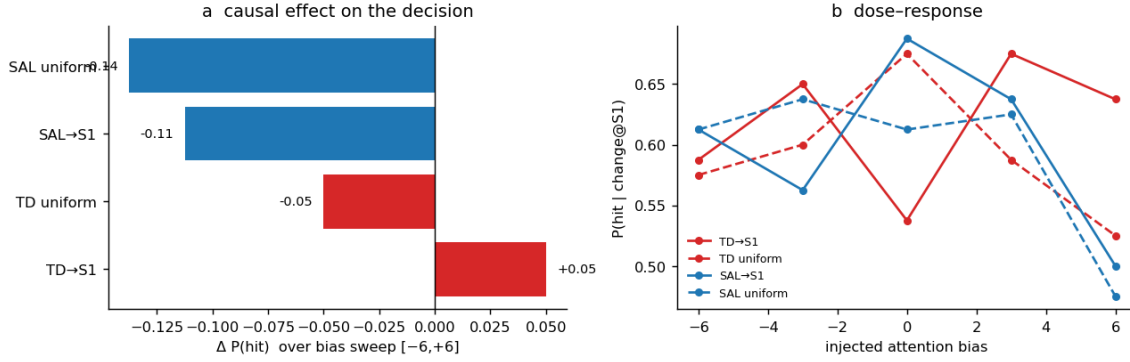


Figure 6: Causal effect of biasing each attention lever. (a) Change in hit rate over the full bias sweep for each lever — biasing top-down attention onto the changed quadrant is the strong decision lever; uniform gains and salience levers are weak. (b) Dose-response: hit rate versus injected bias for each lever.

of correctly waiting for the end-of-trial reward. This is a graded, not binary, value representation, even though the policy in §3.1 thresholds it into near-binary behaviour (acting for red and green, abandoning blue).

The decision signal is value-gated in time. The press-minus-wait advantage is negative on every plateau before the change and jumps toward press exactly at the change frame, but only for valued trials: it crosses to positive for red ($-0.33 \rightarrow +0.04$), rises for green ($-0.30 \rightarrow -0.07$), and stays negative for blue (-0.23 , abandon) and for no-change (-0.18 , keep waiting). Finally the critic is calibrated to the evidence: its outcome uncertainty just after the change falls monotonically as the change magnitude grows, from an entropy of 0.52 at the smallest change to 0.10 at 44° (with a slight uptick at the extreme). More detectable changes leave the agent more certain of the trial's outcome.

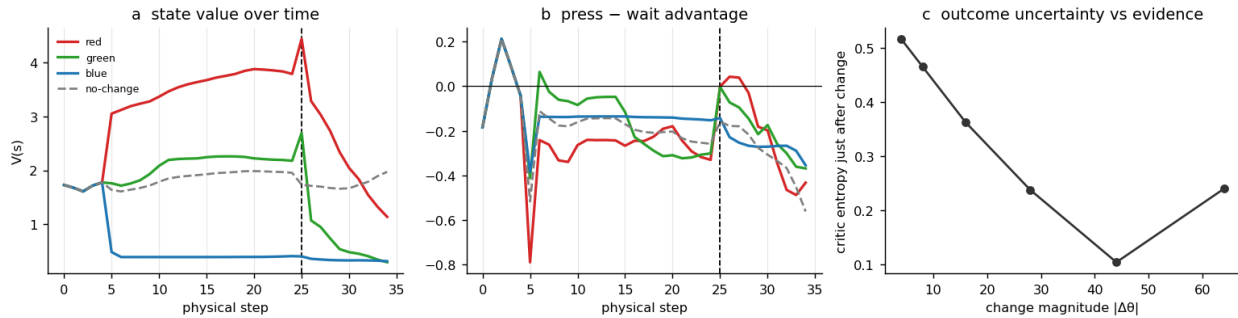


Figure 7: The distributional critic under a forced-wait policy. (a) State value over time by cue colour — high-value colours rise together, the lowest-value colour is discounted, and no-change trials sit at an intermediate value. (b) The press-minus-wait advantage jumps positive within the change frame for valued trials. (c) Outcome uncertainty (critic entropy just after the change) falls as the change magnitude grows.

4. Discussion

4.1 A grounded stream and a triggered gate

The held-frame protocol does not change the architecture, but it makes its computation legible. The two streams separate cleanly into a grounded salience read — stable across plateaus, mildly

cue-shaped, present from the start — and a change-triggered top-down gate that is flat until the change and then jumps onto the changed location. Decoding (§3.3) and causal injection (§3.4) agree with the attention maps (§3.2): the spatial identity of the change is created at the change and held in the top-down memory, and it is the top-down stream — not the salience stream — whose bias onto that location moves the decision. This is the same qualitative division of labour found for this architecture on the standard un-held task, recovered here on a thirty-five-step held timeline where the dynamics are visibly stepwise.

4.2 Why holding frames helps, and what the plateaus reveal

Holding each frame for five identical steps lets this conv-free agent learn a task it otherwise struggles to learn: the frozen, repeated percept gives the recurrent state several steps to settle before it must act, and detection accuracy — not reaction time — is what improves, since the agent presses on the first held step once it detects. Viewed as dynamics, the trajectory is a staircase: the state relaxes onto a plateau within each held frame and jumps when the frame advances, most sharply at the change. The top-down gate’s S1 read is the clearest instance — a flat plateau near 0.31 followed by a jump to ~ 0.48 — and the changed-quadrant code switches from chance to near-ceiling across the same single step. This is exactly the plateau-and-jump picture that motivates treating the working-memory state as a point that drifts on basins and jumps between them.

4.3 Representation matches use

The agent’s internal code is selective in a telling way. It maintains the cue colour, which sets the trial’s value and which it acts on, at ceiling for the whole trial; it perceives the cue reliability but lets it decay to chance, and its behaviour ignores reliability entirely (flat psychometric across rings, no validity benefit). The critic, meanwhile, represents value as a graded quantity ordered by reward, even though the policy collapses that graded value into near-binary action. What the agent keeps and what it discards line up with what its policy needs — value in, an unused spatial-reliability cue out.

4.4 Limitations

The causal decision effect of the top-down lever, while the only positive one, is modest and close to the trial-to-trial noise floor on this conservative detector; we therefore present it as directional support for the decoding and attention results rather than as a strong stand-alone lever. The agent’s hit rate on change trials is moderate (≈ 0.46) by design of its conservative criterion, so the psychometric curves saturate below ceiling. Reaction time is uninformative under the held protocol because detection is effectively immediate within the change frame; a finer-grained chronometric analysis would require single-step (un-held) frames. Finally, this is a single trained agent of one architecture; the cross-architecture comparison is the subject of the broader program.

5. Conclusion

A parallel dual-stream recurrent vision transformer, trained by reinforcement on cued, value-weighted change detection with each frame held for five steps, solves the task as a conservative, well-calibrated, value-directed detector with no spatial cueing benefit. The held-frame timeline exposes a clean separation between a grounded salience stream that holds the scene and a change-triggered top-down gate that locks onto the changed location and carries the decision,

with a critic whose graded value code is calibrated to the strength of the evidence. The agent maintains the cue information it uses and discards the information it ignores, and its internal dynamics take the form of plateaus punctuated by jumps at the moments that matter.